# What seniors no longer teach: learning efficiency, intergenerational knowledge transmission and the AI skill cliff. An agent-based approach

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**Abstract**

In the growth model of Lucas (1988) the efficiency with which workers accumulate human capital, $B$, is a constant. In the later work of Lucas himself (Lucas, 2009; Lucas and Moll, 2014), knowledge is acquired in encounters between people who know more and people who know less. Recent evidence that generative AI reduces the employment of entry-level workers (Brynjolfsson et al., 2025; Hosseini and Lichtinger, 2025) turns this distinction into a policy question: if AI substitutes for junior work, it may also remove the occasions on which junior workers learn from senior ones (Ide, 2026). This paper develops an agent-based model in which $B$ is not assumed but emerges from junior–senior meetings, and uses it to characterise the timing of the resulting losses. Calibrated to an age–wage profile that rises slowly and flattens, as in the Italian private sector, the model shows that a permanent halving of meeting opportunities leaves the human capital of senior workers *exactly* unchanged for ten years and erodes it thereafter, through a cumulative and intergenerational mechanism, by 6.7 per cent in the long run. We call this delayed fall the *skill cliff*. Introducing AI explicitly, we find that who is substituted matters: removing tasks from juniors or from seniors lowers aggregate human capital by the same 5.2 per cent after fifty years but with different effects on inequality, seniors who lose their jobs matter only when mentoring capacity is scarce, and with a conservative estimate of AI productivity the loss of human capital absorbs about 86 per cent of the output gain. Two propositions connect the micro-level rules to the aggregate: an

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identity that expresses $B$ as a function of meeting opportunities, mentoring capacity and the distribution of knowledge, and a transmission lag equal to the distance between the junior and senior thresholds plus one year.

*Keywords:* Human capital, Knowledge diffusion, Artificial intelligence, Agent-based model, Mentoring, Skill cliff
*JEL:* C63, J24, O33, O41, J11

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## 1. Introduction

In 1988 an economist from Chicago wrote an equation that changed how economists think about the development of nations. Solow (1956) had explained growth through the accumulation of physical capital and a technology that fell from outside the model like rain. Lucas (1988) brought growth inside: it originates in the decisions of agents about how to use their time, and its engine is human capital, the knowledge embodied in people. In his model human capital grows as

$$\dot{h} = B\,(1-u)\,h, \tag{1}$$

where $u$ is the share of time devoted to production, $1-u$ the share devoted to learning, and $B$ the efficiency of learning: how much new human capital is generated by each hour spent learning. In 1988, $B$ is a constant. Lucas fixes it; he does not explain it.

The choice was a necessary simplification, and it has produced insights that still stand. But it also hides a question that has become urgent: what does $B$ depend on? Two people who devote the same time to learning, starting from the same human capital, will learn at very different speeds if the context differs. A resident in a teaching hospital, surrounded by complex cases and experienced surgeons, has a higher $B$ than a doctor working alone in a small practice. A junior programmer whose code is reviewed by a senior engineer, who explains the mistakes before they matter, has a higher $B$ than one whose only interlocutor is a search engine. $B$ measures the quality of the context in which one learns, and that context is made of people.

Lucas himself moved in this direction. In Lucas (2009), growth emerges from individuals who learn by observing and imitating those who do better; in Lucas and Moll (2014), agents divide their time between producing with the knowledge they have and searching for encounters with others from whom to learn, and what a learner gains depends on the distance between what the learner knows and what the counterpart knows. Empirical work

on coworker learning supports this view: wage growth is faster for workers whose colleagues earn more, and the value of this learning amounts to 4 to 9 per cent of compensation (Jarosch et al., 2021; Herkenhoff et al., 2024).

The distinction is not merely formal. If $B$ is a parameter, a technology that changes the organisation of work leaves the efficiency of learning untouched. If $B$ emerges from meetings, a technology that reduces the number of meetings reduces $B$ itself. Generative AI provides a pressing case. Early evidence suggests that it reduces the employment of young and entry-level workers in exposed occupations, while leaving senior employment broadly unchanged (Brynjolfsson et al., 2025; Hosseini and Lichtinger, 2025). Ide (2026) shows, in an overlapping-generations model, that automating entry-level tasks makes seniors more productive today but cuts the channel through which tacit knowledge passes to the next generation; because contracts for the transfer of tacit knowledge are inherently incomplete, the market automates more than is socially optimal, and long-run growth per capita may fall by 0.05 to 0.35 percentage points a year.

The literature is, however, lacking in three respects. First, learning efficiency is typically treated as a parameter, even where its microfoundations are explicitly social. Second, the effects of AI on the labour market are measured at impact—on the employment of young workers today—while the transmission channel operates with a delay of one professional generation: nobody can yet observe the human capital of the seniors of 2040. Third, the timing of this delay, and the conditions under which a gradual reduction in meetings turns into a fall in aggregate human capital, have not, to our knowledge, been characterised.

We therefore develop an agent-based model (ABM) in which $B$ is an emergent property of junior–senior meetings, in the spirit of the use of agent-based methods to microfound aggregate dynamics in Bassi et al. (2022) and Railsback and Grimm (2011). The contribution of the paper is fourfold. First, we show that a parsimonious model with demography, meetings and life-cycle learning reproduces a concave age–wage profile and a share of learning from coworkers consistent with Jarosch et al. (2021). Second, we show that a permanent reduction in meeting opportunities produces a *skill cliff*: senior human capital is exactly unaffected for $s_S - s_J$ years, where $s_J$ and $s_S$ are the junior and senior experience thresholds, and declines thereafter as cohorts trained under fewer meetings become the teachers of the next ones. Third, we derive an aggregate identity for $B$ that can be used to make learning efficiency endogenous in single-equation growth models such as Eq. (1). Fourth,

introducing AI explicitly, we show that what substitution means matters as much as whom it targets: removing tasks from juniors (fewer meetings) or from seniors (less effective meetings) costs the same human capital, but the two differ in inequality and in the transitional output gain, which depends on how familiar with AI its users are; that the race between AI productivity and the erosion of human capital can be lost even when AI raises output in the short run, and that firms which do not internalise the transmission of knowledge automate more than is socially optimal.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical background. Section 3 presents the model as implemented. Section 4 reports the baseline simulations and a mechanism experiment that isolates the meeting channel. Section 5 derives the implications for aggregate modelling. Section 6 introduces AI. Section 7 discusses policy and concludes.

## 2. Theoretical background and motivation

### *2.1. Learning efficiency: from parameter to process*

In Lucas (1988) the parameter $B$ pins down the balanced growth rate of the economy and, along the saddle path, the speed at which the economy approaches it. Life-cycle models of investment in human capital (Ben-Porath, 1967) explain why individual learning declines with age, and hence why empirical age–earnings profiles are concave (Murphy and Welch, 1990; Lagakos et al., 2018); but they too treat the productivity of learning as given. In Lucas and Moll (2014) the parameter is replaced by a process: the distribution of knowledge in the population is the state variable, and growth depends on how often people meet, on whom they meet, and on whether the difference in knowledge between them is large enough to make the meeting fruitful. $B$ becomes the endogenous outcome of the structure of encounters among workers: high when juniors frequently meet more capable seniors, low when those encounters disappear.[1]

### *2.2. Automation, entry-level work and the apprenticeship channel*

Tacit knowledge differs from codified knowledge: it is not in manuals and is not transmitted by a slide deck, but acquired through situated practice,

[1]The mapping between the evidence and our measure is approximate: Jarosch et al. (2021) measure learning from all coworkers as a share of the compensation of all workers, while we measure learning of junior workers from senior ones as a share of junior wages.

error and correction in real time. Ide (2026) identifies two forces through which AI erodes its transmission. On the supply side, automating entry-level tasks reduces the apprenticeship opportunities that experts offer. On the demand side, AI co-pilots partly compensate for missing mentors, but they also reduce the effort that novices put into learning by practice: they deliver the output without building the capability—and capability is exactly what is needed to become the senior who will teach the next generation.

The available evidence measures only the first-round effect. Employment of 22–25 year-olds in AI-exposed occupations has fallen by 13–16 per cent relative to older workers (Brynjolfsson et al., 2025), and firms adopting generative AI hire fewer junior workers (Hosseini and Lichtinger, 2025). The second-round effect, on the human capital of future seniors, is by construction not yet observable. A model is needed to discipline expectations about it.

### *2.3. Substitution, complementarity and direction*

Acemoglu (2024) estimates modest aggregate productivity gains from AI over the next decade and argues that its direction matters: AI can be used to augment workers or to replace them, and the industry is choosing the second option because it lowers labour costs in the short run (see also Acemoglu and Restrepo, 2019). In the framework of Eq. (1), the distinction has precise consequences: complementary AI may raise $B$ by making learning more efficient, while substitutive AI lowers it by emptying the contexts in which learning happens. Acemoglu is right that direction matters, but the analysis stops at the short run and at directly automatable tasks: it does not capture the intergenerational mechanism. The problem is not only how many jobs disappear today, but how much tacit knowledge is not transmitted tomorrow.

The connection between these three strands—Lucas and Moll on the social nature of $B$, Ide on automation as a destroyer of encounters, Acemoglu on the direction of technical change—is our interpretation rather than a result of any of them. The model in the next section is designed to test whether it holds together quantitatively.

## 3. The model

### *3.1. Agents and demography*

The economy is populated by $N = 5{,}000$ workers indexed by $i$; each period corresponds to one year. Worker $i$ is characterised by age $a_{i,t}$, years

of experience $s_{i,t}$, a skill level $q_i \in \{L, H\}$ and human capital $h_{i,t}$. At the beginning of each period all workers age by one year, $a_{i,t} = a_{i,t-1} + 1$ and $s_{i,t} = s_{i,t-1} + 1$; those reaching the retirement age $\bar{a} = 65$ leave and are replaced one-for-one by entrants, so that $N$ is constant.[2] An entrant is high-skilled with probability $q = 0.3$, enters with $s = 0$ at age 25 if high-skilled and 20 otherwise, and draws

$$h_{i,0} = \bar{h}_0(q_i)\, e^{\varepsilon_i}, \qquad \varepsilon_i \sim \mathcal{N}\left(-\tfrac{\sigma^2}{2}, \sigma^2\right), \tag{2}$$

with $\bar{h}_0(L) = 1$, $\bar{h}_0(H) = 1.5$ and $\sigma = 0.2$. Roles are defined by experience rather than age: worker $i$ is *junior* if $s_{i,t} < s_J$ and *senior* if $s_{i,t} \geq s_S$, with $s_J = 5$ and $s_S = 15$.

### *3.2. Meetings*

Junior workers learn from seniors of their own skill group $g \in \{L, H\}$: one learns from those who do the same job. Let $J_{g,t}$ and $S_{g,t}$ denote the number of junior and senior workers in group $g$. Each senior can follow at most $\kappa$ juniors per year, so that the number of meetings is $\min(pJ_{g,t}, \kappa S_{g,t})$ and the probability that a junior meets a senior is

$$p^{\text{eff}}_{g,t} = \min\left(p,\ \kappa \frac{S_{g,t}}{J_{g,t}}\right). \tag{3}$$

A junior $j$ who meets a senior $k$, drawn uniformly among the seniors of the same group, gains

$$\Delta h^{\text{inc}}_{j,t} = \beta \max\left(0,\ h_{k,t} - h_{j,t}\right), \tag{4}$$

as in Lucas and Moll (2014): the gain is proportional to the knowledge gap, and nothing is unlearned from a less knowledgeable senior. The senior bears no cost.

### *3.3. Autonomous learning and obsolescence*

All workers also learn autonomously, at a rate that declines with experience as in Ben-Porath (1967), and lose a fraction $d$ of their knowledge to obsolescence:

$$\Delta h^{\text{aut}}_{i,t} = \left[\delta(s_{i,t}) - d\right] h_{i,t}, \qquad \delta(s) = \delta_0 \max\left(0,\ 1 - \frac{s}{S_h}\right). \tag{5}$$

[2] A variant with the Italian age pyramid, in which $N$ varies over time, is left to a dedicated case study.

Both components are computed on beginning-of-period human capital and applied simultaneously, $h_{i,t+1} = h_{i,t} + \Delta h^{\text{inc}}_{i,t} + \Delta h^{\text{aut}}_{i,t}$; $h$ remains non-negative because $d < 1$.

### *3.4. Output, wages and emergent learning efficiency*

Output is $Y_t = \sum_i h_{i,t}$ and wages equal marginal products, $w_i = h_i$.[3] Emergent learning efficiency is measured as the mean growth rate of junior human capital,

$$B_t = \underbrace{\frac{1}{J_t}\sum_j \frac{\Delta h^{\text{inc}}_{j,t}}{h_{j,t}}}_{B^{\text{inc}}_t} + \underbrace{\frac{1}{J_t}\sum_j \frac{\Delta h^{\text{aut}}_{j,t}}{h_{j,t}}}_{B^{\text{aut}}_t}, \tag{6}$$

which corresponds to $\dot{h}/h = B(1-u)$ in Eq. (1) with the time devoted to learning normalised to one.

### *3.5. Calibration and implementation*

Table 1 reports the parameter values. Demographic parameters reflect the Italian labour market; $\beta$ is chosen so that $B^{\text{inc}}$ falls within the range estimated by Jarosch et al. (2021); $(\delta_0, S_h, d)$ are chosen to produce an age–wage profile that rises slowly, peaks late and declines only mildly, as in the Italian private sector. The model is implemented in vectorised Python; each run includes a burn-in of 300 years, and statistics are averaged over 30 replications with independent seeds and 95% confidence intervals. Scenarios share random numbers replication by replication. An independent re-implementation of the same rules with one object per agent in Mesa yields results that are statistically equivalent within ±1 per cent (Appendix Appendix B), a form of model docking (Axtell et al., 1996). A complete description following the ODD protocol is given in Appendix Appendix A. Code and data are available at `https://github.com/ViciusLio/skill-cliff-abm`.

## 4. Simulation results

### *4.1. Baseline: stylised facts*

The baseline reproduces a concave age–wage profile (Figure 1). Wages peak after 34–35 years of experience in both skill groups, at a level 1.84

[3] Output becomes $Y_t = f(\sum_i h_{i,t}, A_t)$ in the extension with AI.

Table 1: Parameters of the baseline model.

| Parameter | Value | Description | Source / criterion |
|---|---|---|---|
| $N$ | 5,000 | workers | — |
| entry age | 20 / 25 | low / high skill | Italian graduates enter late |
| $\bar{a}$ | 65 | retirement age | effective exit age, Italy 2024 |
| $q$ | 0.30 | share of high-skilled entrants | tertiary share, age 25–34 |
| $\bar{h}_0(L), \bar{h}_0(H)$ | 1.0 / 1.5 | mean entry human capital | normalisation |
| $\sigma$ | 0.20 | dispersion of entry $h$ | arbitrary |
| $s_J$ / $s_S$ | 5 / 15 | junior / senior thresholds | entry-level evidence |
| $p$ | 0.60 | meeting probability | arbitrary (with $\beta$) |
| $\kappa$ | 0.20 | mentoring capacity | arbitrary |
| $\beta$ | 0.10 | meeting effectiveness | $B^{\text{inc}} \in [4\%, 9\%]$ |
| $\delta_0, S_h, d$ | 0.04 / 50 / 0.012 | autonomous learning, obsolescence | concave profile |

times the entry wage, and decline by 3.6 (low-skilled) and 0.8 (high-skilled) per cent before retirement. Emergent learning efficiency is $B = 0.0693$ (95% CI $[0.0692, 0.0694]$), of which $B^{\text{inc}} = 0.0429$, or 62 per cent, comes from meetings (Figure 3, panel a). The ratio of seniors to juniors is 5.71, so that mentoring capacity does not bind in the steady state ($p^{\text{eff}} = p$).

Two discrepancies should be noted. First, the profiles of the two skill groups have the same slope, while the evidence indicates that they are steeper for the more educated (Lagakos et al., 2018); reproducing this would require a skill-specific autonomous learning rate, which we leave out to keep the model essential. Second, the wage Gini coefficient (0.155) is well below observed values, because human capital is the only determinant of wages in the model. Neither is a calibration target.

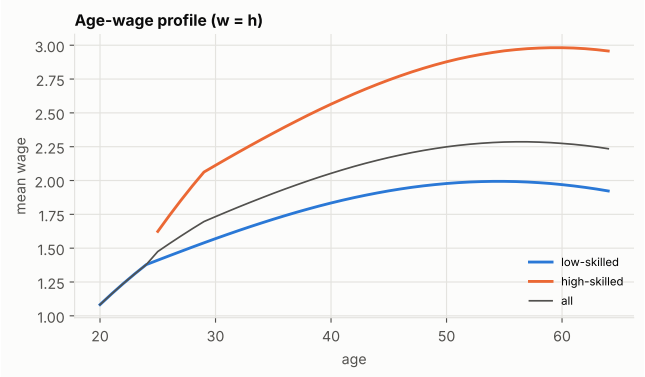



Figure 1: Age–wage profile in the baseline. Mean over 30 replications and 60 years; bands are 95% confidence intervals.

### *4.2. The skill cliff: a mechanism experiment*

To isolate the channel through which AI will operate, we halve $p$ permanently from year 10 onwards, holding everything else constant.[4] Figure 2 reports mean human capital by age class. The response is immediate for junior workers, whose human capital is 5.7 per cent lower after four years. For senior workers the response is delayed: their mean human capital is *exactly* unchanged for ten years, starts falling in the eleventh, is 3.6 per cent lower after 25 years and 6.7 per cent lower in the new steady state, while output falls by 7.2 per cent. The loss travels along the age distribution like a wave: it exceeds 0.5 per cent at once for workers aged 20–29, and after about 4, 14, 24 and 33 years for the following age classes.

The path of learning efficiency is non-monotonic (Figure 3, panel b). $B^{\text{inc}}$ halves at impact, from 0.043 to 0.021; it then rises to 0.025 over the following two decades, because juniors start from lower levels and the gap to their seniors widens; and it falls again, to 0.020 after 50 years, as seniors trained under fewer meetings become the teachers. Learning efficiency inherits the history of meetings: it is not a parameter but a state variable. Finally, the wage Gini rises from 0.155 to 0.165 (Figure 4): meetings compress inequality by pulling juniors towards seniors.

[4] This is not an AI scenario: it is the comparative dynamics of the meeting channel alone.

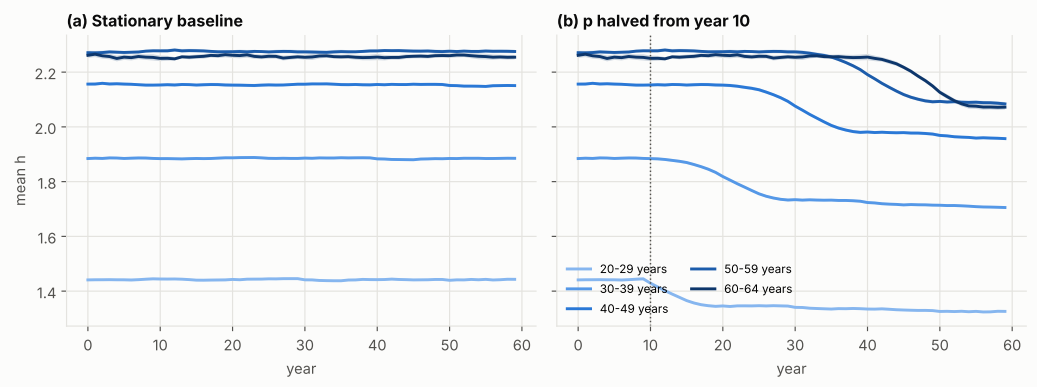



Figure 2: Mean human capital by age class. (a) Baseline. (b) Meeting probability halved from year 10. Mean over 30 replications; bands are 95% confidence intervals.

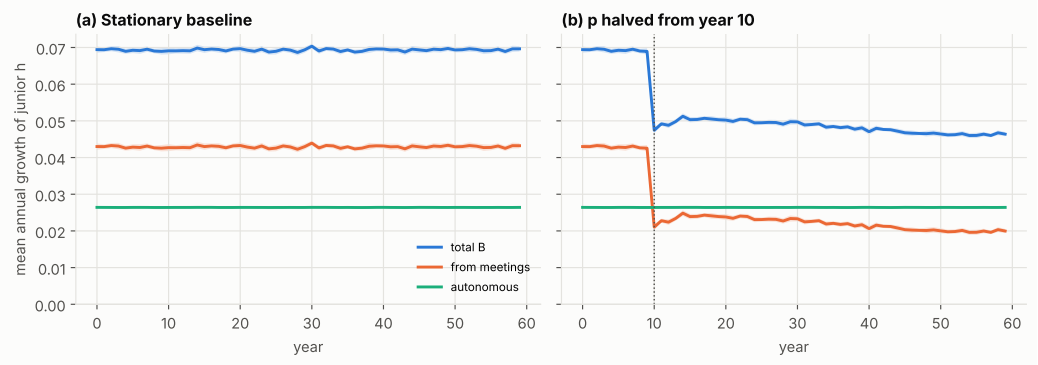



Figure 3: Emergent learning efficiency of junior workers and its decomposition.

### *4.3. Mentoring capacity and robustness*

Figure 5 shows $B^{\text{inc}}$ and mean senior human capital as functions of $p$ for different values of $\kappa$. Both increase more than linearly in $p$, reflecting the cumulative mechanism described above, until the capacity constraint in Eq. (3) binds; beyond that point additional demand for meetings is rationed. In the baseline the constraint becomes active only if $S/J$ falls below $p/\kappa = 3$, that is, after a large reduction in the number of seniors relative to juniors.

With thresholds $s_J = 10$ and $s_S = 20$, $B$ falls to 0.057 because the junior pool includes workers further from entry, but the long-run effects of halving $p$ are similar ($-6.6$ per cent for senior human capital, $-7.2$ per cent for output) and the transmission lag is again 11 years: it depends on the distance between the thresholds, not on their level.

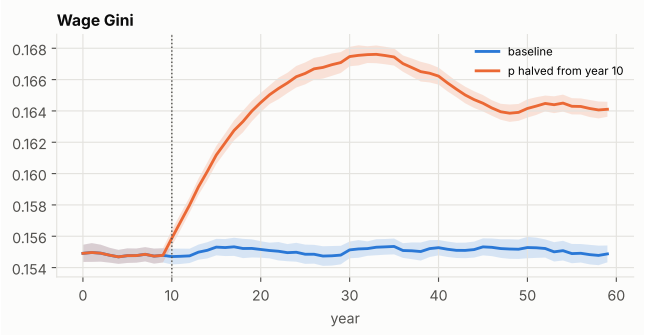



Figure 4: Wage Gini coefficient.

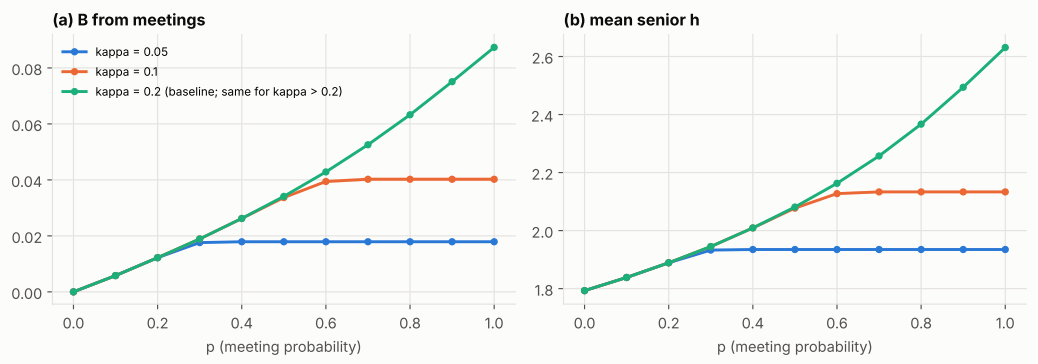



Figure 5: Sensitivity to the meeting probability $p$ and to mentoring capacity $\kappa$ (10 replications per point).

## 5. Implications for aggregate modelling

Single-equation growth models remain popular, and with good reason. Are there lessons from the agent-based model that can be incorporated into them? We argue that there are two.

**Proposition 1** (Emergent learning efficiency)**.** *Let $\bar{h}^S_{g,t}$ be mean senior human capital in group $g$ and $\overline{h^{-1}}^J_{g,t}$ the mean of the inverse of junior human capital in the same group, both at the beginning of period $t$. Under random matching within groups, the expected contribution of meetings to learning efficiency is*

$$\mathbb{E}\left[B^{\text{inc}}_t\right] = \sum_g \frac{J_{g,t}}{J_t}\, \beta\, p^{\text{eff}}_{g,t} \left[\bar{h}^S_{g,t}\, \overline{h^{-1}}^J_{g,t} - 1 + \Omega_{g,t}\right], \tag{7}$$

*where*

$$\Omega_{g,t} = \frac{1}{J_{g,t} S_{g,t}} \sum_{j,k \in g} \frac{\max(0, h_{j,t} - h_{k,t})}{h_{j,t}} \geq 0.$$

*Proof.* Since $\max(0, x) = x + \max(0, -x)$, the expected gain of junior $j$ from a meeting with a uniformly drawn senior of the same group is $\beta(\bar{h}^S_{g,t} - h_{j,t}) + \beta S^{-1}_{g,t} \sum_{k \in g} \max(0, h_{j,t} - h_{k,t})$. Dividing by $h_{j,t}$, multiplying by $p^{\text{eff}}_{g,t}$ and averaging over juniors gives the result. □

Eq. (7) is the aggregate counterpart of the micro-level rules. Learning efficiency depends on three objects that a constant $B$ suppresses: meeting opportunities ($p$), mentoring capacity relative to demand ($\kappa S/J$), and the distribution of knowledge between generations ($\bar{h}^S \overline{h^{-1}}^J$). In the baseline the identity is verified numerically (0.0437 against a simulated 0.0434). The dispersion of junior human capital matters: replacing $\overline{h^{-1}}^J$ with $1/\bar{h}^J$ underestimates $B^{\text{inc}}$ by about 9 per cent.

**Proposition 2** (Transmission lag)**.** *Suppose that meeting opportunities change permanently at $t_0$ and that seniors do not learn from meetings. Then mean senior human capital is unaffected for $t \leq t_0 + s_S - s_J$, and can be affected from $t_0 + s_S - s_J + 1$ onwards.*

*Proof.* A worker who is junior at $t_0$ has experience at most $s_J - 1$ and becomes senior when their experience reaches $s_S$, that is, after at least $s_S - s_J + 1$ years. Until then the set of seniors consists of workers whose junior years ended before $t_0$, and whose human capital evolves according to Eq. (5) alone. □

With $s_J = 5$ and $s_S = 15$, senior human capital is unchanged for ten years and the first effect appears in the eleventh, exactly as observed in the simulations. Proposition 2 explains why the first-round evidence on young workers (Brynjolfsson et al., 2025) cannot, by itself, reveal the second-round effect on human capital: the latter is invisible for a full professional generation, and then unfolds over several. The long-run elasticity of senior human capital to meeting opportunities, $\varepsilon = \Delta \ln \bar{h}^S_\infty / \Delta \ln p \approx 0.10$ in the baseline, plays here the role that the degree of hysteresis plays in aggregate models of the normal rate of capacity utilisation (Bassi et al., 2022): it summarises in a single number a property that emerges from heterogeneous micro-behaviour.

## 6. Introducing artificial intelligence

### *6.1. Specification*

AI arrives in year $t_0 = 10$ of the recorded period. Its productivity grows at rate $g$, and it automates a growing share $\varphi_t$ of the tasks of the targeted workers:

$$A_t = (1+g)^{t-t_0} \quad (t \geq t_0), \qquad \varphi_t = \varphi_{\max}\left(1 - \frac{1}{A_t}\right), \tag{8}$$

$$Y_t = H_t\left[1 + \theta\left(A_t - 1\right)F_t\right], \tag{9}$$

with $A_t = 1$ before $t_0$ and $H_t = \sum_i h_{i,t}$. The share $\varphi_t$ starts at zero and saturates at $\varphi_{\max}$.

The gain from AI depends on how familiar with it are the workers who use it, $U_t$: the workers who are not targeted (non-juniors if AI replaces juniors, non-seniors if it replaces seniors, the targeted skill group in the skill-based scenarios, everyone for complementary AI). Each worker has a familiarity $f_i \in [0,1]$: workers who enter after $t_0$ are trained with AI and have $f_i = 1$; incumbents start from $f_i = 0$ and, in each year of employment with AI, update $f_i \leftarrow f_i + \gamma(1 - f_i)$, with $\gamma = 0.1$. $F_t = \sum_{i\in U} h_i f_i / \sum_{i\in U} h_i$ is the human-capital-weighted familiarity of users. Users earn $w_i = h_i[1 + \theta(A_t - 1)f_i]$ and the other workers $w_i = h_i$; the remainder of output, $\theta(A_t - 1)F_t(H_t - \sum_{i\in U} h_i)$, is a rent that accrues to the owners of AI. The level of the gain is thus proportional to the whole of human capital, including that embodied in the tasks that AI performs instead of the targeted workers, while its users determine how effectively it is exploited; with $f_i = 1$ for all workers Eq. (9) reduces to the form used without familiarity, so that $\theta$ keeps its economy-wide interpretation. Familiarity does not affect learning, so that it changes output and wages but not the dynamics of human capital; since every worker has entered after $t_0$ by the end of the horizon, $F_t = 1$ fifty years after the introduction of AI.

What does it mean to "substitute" a worker? We distinguish the target (juniors or seniors) from the margin: the worker keeps the job but loses tasks, or loses the job. The four cases act on different parameters of the meeting technology (Table 2). Replacing junior tasks multiplies $p$ by $1 - \varphi_t$ (fewer occasions to work alongside a senior). Replacing junior jobs makes a share $\varphi_t$ of juniors non-employed each year: they neither meet seniors nor learn on the job, their human capital depreciates at rate $d$, and the year does not count as experience, so that they remain juniors longer and reach seniority later.

Table 2: Four meanings of substitution.

| | Loses tasks | Loses the job |
|---|---|---|
| Junior | $p \times (1 - \varphi_t)$: fewer meetings | non-employed with probability $\varphi_t$: no meetings, no experience |
| Senior | $\beta \times (1 - \varphi_t)$: less transmitted per meeting | $\kappa \times (1 - \varphi_t)$: fewer mentors |

Their tasks are performed by AI, so that current output is unaffected and the loss operates only through human capital. Replacing senior tasks multiplies $\beta$ by $1 - \varphi_t$ (each meeting transmits less); replacing senior jobs multiplies $\kappa$ by $1 - \varphi_t$ (fewer seniors available as mentors). Replacing high- or low-skilled workers applies the junior and the senior cut within the corresponding group, and complementary AI multiplies $\beta$ by $1+\varphi_t$ (AI makes teaching more effective). Task substitution is the default for both roles.

We set $g = 0.03$ and $\varphi_{\max} = 0.6$, so that ten years after its introduction AI automates 15 per cent of the targeted tasks, in line with the 13–16 per cent relative fall in the employment of young workers in exposed occupations (Brynjolfsson et al., 2025); and $\theta = 0.02$, which yields a productivity gain of 0.7 per cent over ten years, close to the conservative estimate of Acemoglu (2024). Since $\theta$ does not affect the dynamics of agents, the output path can be computed for any $\theta$ from $H_t$, $A_t$ and $F_t$. Every scenario shares random numbers with the scenario without AI, so that before $t_0$ the two are identical and all differences are due to AI. We use $N = 50{,}000$ and 30 replications.[5]

### *6.2. Who is substituted matters*

Table 3 and Figure 6 compare the scenarios fifty years after the introduction of AI. Replacing junior tasks lowers aggregate human capital by 5.2 per cent and almost halves learning from meetings. Consistently with Proposition 2, seniors are affected only twelve years after the introduction of AI—one year for $A_t$ to exceed one, plus the eleven-year transmission lag. Replacing senior tasks has the same mean effect: because $p$ and $\beta$ enter learning as a

[5]The model is invariant to scale: because meetings are random draws within large pools rather than links in a local network, simulations with 5,000 and with 24 million agents (the number of employed workers in Italy) yield the same results to the fourth digit; the population size only reduces sampling noise.

product (Eq. (7)), cutting either by $1 - \varphi_t$ yields the same loss of human capital (the paired difference is 0.002 percentage points, with a 95 per cent confidence interval that includes zero). The two channels differ in distribution and timing. Fewer meetings mean that some juniors learn and others do not, whereas less effective meetings mean that every junior learns a little less: the Gini index of human capital is 0.163 in the first case and 0.161 in the second. Who uses AI matters more: when it replaces junior tasks its gain accrues to experienced workers and the wage Gini rises to 0.167, when it replaces senior tasks the gain accrues to less experienced workers and the wage Gini stays at 0.155, as without AI. The users of AI that replaces seniors have entered after its introduction and are more familiar with it, so that output is higher in the transition: ten years after the introduction it exceeds the no-AI level by 0.35 per cent, against 0.22 per cent when AI replaces juniors. The advantage vanishes as the incumbents retire, and is small with $\theta = 0.02$: the trade-off between output today and human capital tomorrow exists, but it is not where the bulk of the skill cliff lies. Replacing low-skilled workers costs more than replacing high-skilled workers and raises wage inequality, because the low-skilled are 70 per cent of entrants; since these scenarios cut both $p$ and $\beta$ within the group, learning in the group scales with $(1 - \varphi_t)^2$. Complementary AI mirrors task substitution.

If juniors lose their jobs instead of tasks, training is postponed rather than lost: since experience accrues only in employment, a non-employed junior remains a junior until five years of work have been completed. The loss of human capital (5.7 per cent) is close to that of task substitution, but its composition differs: entry into seniority is delayed and careers are shorter, so that after fifty years there are 11 per cent fewer seniors, while the mean human capital of those who reach seniority falls by only 1.9 per cent.

Seniors who lose their jobs are almost harmless in the baseline, but only because mentoring capacity is abundant: with $S/J \approx 5.7$ there are more potential mentors than needed. When capacity is scarcer ($\kappa = 0.1$), the same substitution lowers aggregate human capital by 4.8 per cent, almost as much as task substitution. How many seniors actually act as mentors is therefore a key empirical quantity.

*Stable mentors.* In the baseline a junior meets a randomly drawn senior every year. In a variant, a junior who meets a senior keeps that mentor for as long as both remain employed and the junior remains a junior, unless the pair separates with annual probability $\lambda$; pairs occupy mentoring capacity, and $\lambda = 1$ reproduces the baseline exactly. With $\lambda = 0.15$, close to the

Table 3: Effects of AI fifty years after its introduction, relative to the scenario without AI (mean of 30 paired replications).

| Target, margin | $H$ | Senior $h$ | $B^{\text{inc}}$ | $Y_{+10}$ | $Y_{+50}$ | Gini |
|---|---|---|---|---|---|---|
| Junior, tasks ($p$) | −5.2% | −4.5% | −45% | +0.22% | +0.9% | 0.167 |
| Junior, jobs | −5.7% | −1.9% | −44% | +0.06% | +0.4% | 0.173 |
| Senior, tasks ($\beta$) | −5.2% | −4.5% | −45% | +0.35% | +0.9% | 0.155 |
| Senior, jobs ($\kappa$) | −0.1% | 0.0% | −2% | +0.62% | +6.4% | 0.149 |
| Low-skilled, tasks | −5.5% | −5.0% | −49% | +0.18% | +0.7% | 0.169 |
| High-skilled, tasks | −3.3% | −2.9% | −21% | +0.30% | +3.0% | 0.153 |
| Complementary | +5.2% | +4.3% | +45% | +0.76% | +12.0% | 0.150 |

$Y_{+10}$, $Y_{+50}$: output ten and fifty years after the introduction of AI, $\theta = 0.02$. Gini: wages at the end of the horizon; 0.155 without AI. Seniors losing their jobs with scarcer mentoring capacity ($\kappa = 0.1$): $H$ −4.8%. For non-employed juniors, $B^{\text{inc}}$ is averaged over all juniors.

annual separation rate of young workers, and $\beta$ recalibrated (from 0.10 to 0.075) to keep $B^{\text{inc}}$ at its Phase 1 value, the age–wage profile, the share of learning from meetings and the wage Gini are virtually unchanged, and the transmission lag of Proposition 2 is still eleven years. Proposition 1 becomes an approximation, which overstates $B^{\text{inc}}$ by about 5 per cent because matched juniors are not a random sample. The symmetry between $p$ and $\beta$ breaks: since $p$ only governs the formation of new pairs, replacing junior tasks lowers aggregate human capital by 3.5 per cent, against 5.3 per cent when senior tasks are replaced; replacing junior jobs costs 7.2 per cent, because a junior who leaves employment also loses the mentor. The variant changes magnitudes, not the timing or the sign of the effects.

### *6.3. The race*

The skill cliff hypothesis is ultimately a race between the growth of AI productivity and the erosion of the knowledge it relies upon. With $\theta = 0.02$ and the junior target, AI alone would raise output by 6.5 per cent after fifty years; with the induced loss of human capital the gain is 0.9 per cent, so that the skill cliff absorbs about 86 per cent of it (Figure 7). The share absorbed is already 55 per cent after ten years and 73 per cent after twenty-five: in the early years the gain from AI is smaller, because incumbents are not yet familiar with it, while the loss of human capital is the same. The break-even value of $\theta$—below which output ends lower than without AI—is 0.017 for

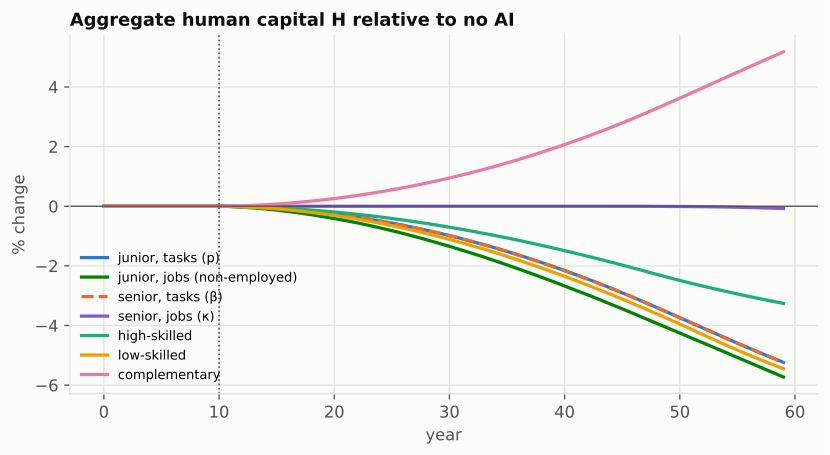



Figure 6: Aggregate human capital relative to the scenario without AI, by target and margin of substitution (the dashed line, senior tasks, almost coincides with junior tasks). Mean over 30 replications; bands are 95% confidence intervals.

$g = 0.03$, and 0.019 if juniors lose their jobs rather than tasks. Faster AI ($g = 0.06$) causes a larger loss of human capital ($-7.5$ per cent) but also a faster growth of $A_t$, so that the break-even value of $\theta$ falls (0.005): with exponential AI productivity the race is eventually won, and the question becomes the horizon over which it is lost.

### *6.4. Private versus social automation*

Why would firms substitute junior workers beyond what is socially desirable? Following Ide (2026), contracts for the transmission of tacit knowledge are incomplete: a firm that trains a junior does not capture the value of the knowledge that the junior will later transmit. We capture this by comparing a firm and a planner who choose a constant share $\varphi$ of automated junior tasks from $t_0$ onwards. Net output relative to no automation is

$$y_t(\varphi) = r_t(\varphi)\left(1 + \pi\varphi - \tfrac{c}{2}\varphi^2\right), \qquad r_t(\varphi) = \frac{H_t(\varphi)}{H_t(0)}, \tag{10}$$

where $\pi = 0.10$ is the gain from full automation and $c = 1/3$ an adoption cost. The firm takes human capital as given and chooses $\varphi = \pi/c = 0.30$; the planner maximises $\sum_k \rho^{k+1} y_{t_0+k}(\varphi)$ over a given horizon, with $r_t(\varphi)$ measured in the model.

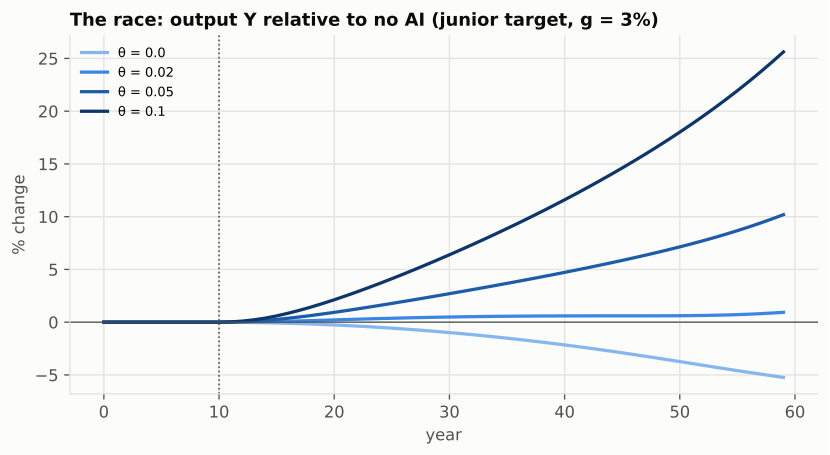



Figure 7: Output relative to the scenario without AI, junior target, $g = 0.03$, for different values of $\theta$.

Figure 8 shows that the firm automates too much, and that the excess grows with the planner's horizon. With $\rho = 0.97$, the optimal share is 0.25 over ten years, 0.19 over twenty-five and 0.09 over fifty. Over fifty years the firm's choice yields a discounted net output 0.6 per cent *below* that without any automation, against +0.14 per cent at the optimum. Over ten years, instead, the two choices almost coincide, because the loss of senior human capital materialises only after the transmission lag of Proposition 2: a decision-maker whose horizon is a balance sheet or an electoral cycle behaves almost like the firm. The values of $\pi$ and $c$ determine the level of the firm's choice, not the sign of the excess, which follows only from the fact that the firm does not see $r_t(\varphi)$.

## 7. Discussion and conclusions

### *7.1. A market failure that nobody sees*

When a firm automates junior work, its accounts improve at once: senior productivity rises and costs fall. No item in the balance sheet records that the next generation of workers will receive less learning, and no economic indicator captures the fall of $B$ that takes place within organisational hierarchies. The structure of the problem resembles pollution: those who pollute obtain an immediate competitive advantage, while those who bear the cost are dispersed in time and space and are often not yet born. Proposition 2 and

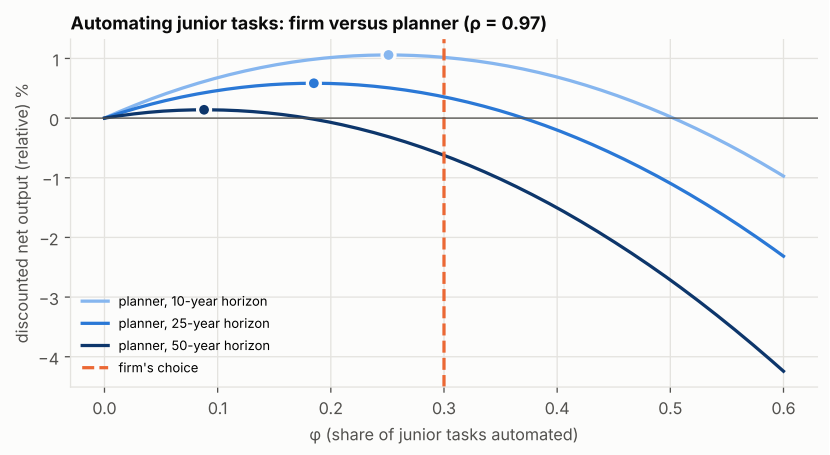



Figure 8: Discounted net output relative to no automation as a function of the automated share of junior tasks, for planners with different horizons ($\rho = 0.97$). Dots mark the planner's optimum; the dashed line marks the firm's choice.

Section 6 add a quantitative dimension to this analogy: in our calibration the cost is invisible in senior human capital for ten years, a firm automates more than three times the share that a planner with a fifty-year horizon would choose, and when the cost appears the policies that could prevent it are already late—the training of a senior takes fifteen to twenty years of exposure to difficult situations, and cannot be accelerated by subsidies or intensive courses.

The responses discussed in the policy debate differ in feasibility and horizon. At the level of the firm, a share of team capacity can be reserved for mentoring and for tasks that juniors own. At the institutional level, apprenticeship programmes can separate training from production and bind seniors and juniors to work together on real problems. At the fiscal level, a contribution on the automation of entry-level tasks would internalise the externality identified by Ide (2026). Finally, AI itself can be directed towards amplifying senior teaching rather than replacing juniors—the direction of technical change that Acemoglu (2024) calls for. In the model these correspond, respectively, to raising $\kappa$, raising $p$, taxing reductions of $p$, and raising $\beta$; Eq. (7) provides a common metric to compare them.

### *7.2. Conclusions*

This paper contributes to the literature in four respects. First, it shows that learning efficiency can be modelled as an emergent property of encounters between workers, and that a parsimonious agent-based model reproduces a concave age–wage profile and a plausible share of learning from coworkers. Second, it shows that a reduction in meeting opportunities produces a delayed and cumulative fall in human capital—a skill cliff—whose timing is governed by the distance between the junior and senior thresholds. Third, it provides an aggregate identity that can be used to make learning efficiency endogenous in single-equation growth models. Fourth, it shows that the effects of AI on human capital depend on whether workers lose tasks or jobs and on whether juniors or seniors are targeted, that a conservative AI productivity gain can be largely absorbed by the skill cliff it causes, and that firms which do not internalise the transmission of knowledge automate too much—by an amount that is invisible over a ten-year horizon.

These contributions aside, the analysis leaves open several avenues for research. Firms are represented by a single decision on the automated share: heterogeneous firms that compete, train and poach workers would make it possible to study how the externality is shared. A demographic variant based on the Italian age pyramid, where large cohorts approach retirement, will make it possible to separate the cliff generated by demography from the one added by AI. Finally, the erosion of skills may interact with the distribution of wealth, if access to the tools of production increasingly takes the form of recurring rents.

The mathematics does not say that $B$ must fall. It says what happens if it is allowed to.

## CRediT authorship contribution statement

**Federico Bassi:** [to be defined]. **Vincenzo Lio:** Conceptualization, Methodology, Software, Writing – original draft.

## Data availability

Code, configuration files and simulated data are available at `https://github.com/ViciusLio/skill-cliff-abm`.

## Appendix A. ODD protocol

This appendix describes the model following the ODD protocol (Overview, Design concepts, Details) as updated by Grimm et al. (2020). It refers to the implementation in the repository (package `skillcliff`); parameter values are those of Table 1 and of the file `base.yaml`.

### *Appendix A.1. Purpose and patterns*

The model studies how the efficiency of learning, $B$, emerges from meetings between junior and senior workers, and how a reduction of those meetings, in particular the substitution of junior work by AI, affects human capital over time. It is evaluated against three patterns: a concave age–wage profile that rises slowly and flattens, as in the Italian private sector; a share of learning from coworkers of 4–9 per cent of junior wages (Jarosch et al., 2021); and a stationary distribution of human capital in the absence of shocks.

### *Appendix A.2. Entities, state variables and scales*

The only agents are workers. Each worker $i$ has an identifier, an age $a_i$ (years), an experience $s_i$ (years of employment since entry), a qualification $q_i \in \{L, H\}$ and human capital $h_i \geq 0$; in the AI scenarios also a familiarity with AI $f_i \in [0, 1]$ and, in the stable-mentor variant, the identifier of the current mentor. The role (junior if $s_i < s_J$, senior if $s_i \geq s_S$, intermediate otherwise) is derived from experience. The global state consists of the time index $t$ and, in the AI scenarios, of AI productivity $A_t$ and the automated share $\varphi_t$. There is no space and no network: interactions are random draws within qualification groups. One time step is one year; a run consists of a burn-in of 300 years, which removes the dependence on initial conditions, followed by 60 recorded years. The population has a constant size $N$ (5,000 in Phase 1, 50,000 in the AI experiments); results are invariant to $N$ up to sampling noise.

### *Appendix A.3. Process overview and scheduling*

In each year the following processes are executed in this order, each on all agents at once (synchronous update):

1. *Ageing*: $a_i$ increases by one; $s_i$ increases by one, and $f_i$ is updated, if the worker was employed in the previous year.
2. *Retirement*: workers with $a_i \geq \bar{a}$ leave.
3. *Entry*: as many workers enter as have left, with $s = 0$.

4. *Roles*: juniors and seniors are identified from experience.
5. *AI* (Section 6 only): the multipliers of $p$, $\kappa$ and $\beta$ for each qualification group are computed from $\varphi_t$; when juniors lose their jobs, a share $\varphi_t$ of the targeted juniors is drawn as non-employed.
6. *Meetings*: within each qualification group, juniors who are employed meet a senior with probability $p_{g,t}^{\text{eff}}$ and compute their gain, Eqs. (3)–(4).
7. *Autonomous learning and obsolescence*: Eq. (5); non-employed juniors only depreciate.
8. *Update*: all gains, computed on beginning-of-year human capital, are added at once.
9. *Observation*: aggregate statistics are recorded (after the burn-in).

*Appendix A.4. Design concepts*

*Basic principles.* Learning efficiency as an outcome of encounters (Lucas and Moll, 2014), life-cycle learning (Ben-Porath, 1967), and substitution versus complementarity of AI (Acemoglu, 2024; Ide, 2026). *Emergence.* Learning efficiency $B$, the age–wage profile, the inequality of wages and the timing of the effects of a reduction of meetings emerge from individual meetings; the transmission lag (Proposition 2) and the dependence of $B$ on the distribution of knowledge (Proposition 1) are emergent properties that we characterise analytically. *Adaptation, objectives, learning, prediction.* Workers follow fixed rules and take no decisions; they do not adapt, optimise or predict. The only decision is the automated share in Section 6, taken outside the agent-based model by comparing simulated paths (firm versus planner). *Sensing.* A junior observes only the human capital of the senior met. *Interaction.* Direct and asymmetric: knowledge flows from the senior to the junior, never in the opposite direction, and only within the same qualification group. Interactions are rationed by the mentoring capacity $\kappa S_g$. *Stochasticity.* The qualification and the initial human capital of entrants, the occurrence of meetings, the choice of the senior and, when juniors lose their jobs, non-employment are random. Each replication uses its own seed, derived from a root seed; separate random streams are used for demography, meetings and AI, so that scenarios with the same seed share all draws until AI arrives (common random numbers). *Collectives.* None beyond qualification groups and roles. *Observation.* Each year: number of juniors, seniors and meetings; effective meeting probability; aggregate human capital $H_t$ and output $Y_t$; mean human capital by role, age and qualification; $B_t$ and its components; the wage

Gini coefficient. Statistics are averaged over 30 replications, with 95 per cent confidence intervals based on Student's $t$.

*Appendix A.5. Initialisation*

$N$ workers are created with a qualification drawn with probability $q$ and an experience drawn uniformly over the career of their qualification group; their age follows from experience and entry age, and their human capital is drawn as for entrants. The burn-in of 300 years makes results independent of this initial state: with 300 instead of 120 years the residual drift of mean human capital falls from $10^{-4}$ to below $10^{-5}$ per year, and the comparisons between scenarios change by less than 0.1 percentage points.

*Appendix A.6. Input data*

The model uses no time series as input. Parameters are read from a configuration file; in the AI scenarios, $A_t$ and $\varphi_t$ are deterministic functions of time, Eq. (8).

*Appendix A.7. Submodels*

*Entry*: Eq. (2), with entry age 20 or 25 by qualification. *Meetings*: Eqs. (3) and (4), applied within each qualification group; the senior is drawn uniformly. *Autonomous learning and obsolescence*: Eq. (5). *Output and wages*: $Y_t = \sum_i h_{i,t}$ and $w_i = h_i$ without AI; Eq. (9) with AI, with wages and familiarity as described in Section 6. *AI*: Eq. (8) and the multipliers described in Section 6; when juniors lose their jobs a targeted junior is non-employed with probability $\varphi_t$, independently each year, and that year does not count as experience. *Firm versus planner*: Eq. (10), evaluated on a grid of automated shares.

*Appendix A.8. Implementation and verification*

The model is written in Python with vectorised NumPy operations (one array per state variable). An automated test suite checks, among other things, that the number of agents is conserved, that human capital is never negative, that $p = 0$ implies no learning from meetings, that runs with the same seed are identical, that every meeting pairs a junior with a senior of the same qualification, and that all AI scenarios coincide with the scenario without AI before $t_0$. An independent object-based implementation in Mesa reproduces the results within ±1 per cent (Appendix Appendix B). Code, configuration and data are available at `https://github.com/ViciusLio/skill-cliff-abm`.

## Appendix B. Docking with an independent implementation

Table B.4 compares the vectorised model with an independent implementation in Mesa (one object per agent), with $N = 5{,}000$ and 30 replications each, as in Phase 1. Two tests are reported for each statistic. The Welch test has the null hypothesis that the two implementations have the same mean: a *large* $p$-value means that no difference is detected. Since failing to detect a difference does not prove equivalence, we also run an equivalence test (two one-sided tests, TOST) with a margin of $\pm 1$ per cent of the mean, chosen to be well below the smallest effect we interpret (about $-2$ per cent): here the null hypothesis is that the difference exceeds the margin, and a *small* $p$-value demonstrates equivalence. The 90 per cent confidence interval of the relative difference shows the narrowest margin within which equivalence holds at the 5 per cent level.

Table B.4: Docking: vectorised (NumPy) vs. object-based (Mesa) implementation ($N = 5{,}000$, 30 replications each).

| Statistic | NumPy | Mesa | Rel. diff. | $p$ (Welch) | 90% CI rel. diff. | $p$ (TOST, $\pm 1\%$) |
|---|---|---|---|---|---|---|
| Mean $h$ | 1.9929 | 1.9920 | $-0.04\%$ | 0.56 | $[-0.17\%, +0.08\%]$ | $< 0.001$ |
| Mean junior $h$ | 1.4184 | 1.4179 | $-0.04\%$ | 0.64 | $[-0.17\%, +0.10\%]$ | $< 0.001$ |
| Mean senior $h$ | 2.1667 | 2.1657 | $-0.05\%$ | 0.55 | $[-0.19\%, +0.09\%]$ | $< 0.001$ |
| $B^{\text{inc}}$ | 0.0429 | 0.0429 | $-0.06\%$ | 0.75 | $[-0.36\%, +0.24\%]$ | $< 0.001$ |
| $B^{\text{aut}}$ | 0.0264 | 0.0264 | 0.00% | 0.32 | $[-0.00\%, +0.00\%]$ | $< 0.001$ |

The stable-mentor variant of Section 6 has been docked in the same way, with the Mesa implementation extended to persistent pairs ($\lambda = 0.15$, $\beta = 0.075$). Equivalence within $\pm 1$ per cent is established for all five statistics ($p_{\text{TOST}} \leq 0.001$); the relative differences lie between $-0.40$ and 0 per cent with 30 replications and between $-0.12$ and $+0.05$ per cent in a second run with 60 replications and a different root seed ($p$ between 0.35 and 0.93 in the Welch test).

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